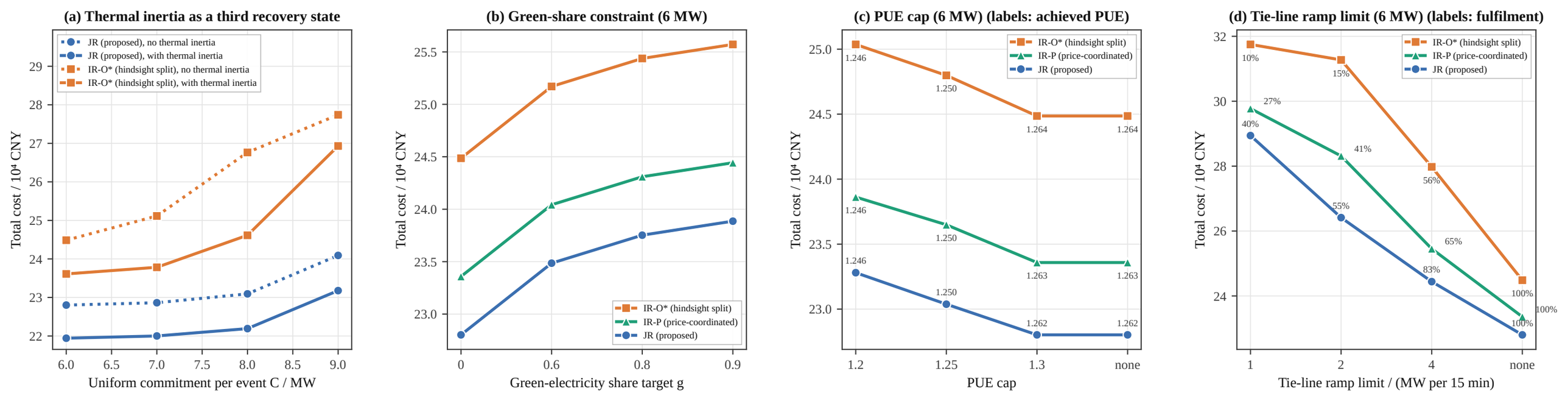
(a) Thermal inertia as a third recovery state
JR (proposed), no thermal inertia
JR (proposed), with thermal inertia
IR-O* (hindsight split), no thermal inertia
IR-O* (hindsight split), with thermal inertia
Total cost / 10⁴ CNY
Uniform commitment per event C / MW
(b) Green-share constraint (6 MW)
IR-O* (hindsight split)
IR-P (price-coordinated)
JR (proposed)
Total cost / 10⁴ CNY
Green-electricity share target g
(c) PUE cap (6 MW) (labels: achieved PUE)
IR-O* (hindsight split)
IR-P (price-coordinated)
JR (proposed)
1.246
1.250
1.264
1.264
1.246
1.250
1.263
1.263
1.246
1.250
1.262
1.262
Total cost / 10⁴ CNY
PUE cap
(d) Tie-line ramp limit (6 MW) (labels: fulfilment)
IR-O* (hindsight split)
IR-P (price-coordinated)
JR (proposed)
10%
15%
56%
100%
27%
41%
65%
100%
40%
55%
83%
100%
Total cost / 10⁴ CNY
Tie-line ramp limit / (MW per 15 min)